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EXAMINER	
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/268,823	Applicant(s) BROOKSHAW et al.	
	Examiner JOHN P GO	Art Unit 3681	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 04 June 2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-7 and 10-16 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-7 and 10-16 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of the Claims

2. **Claims 1-7 and 10-16** are currently pending.

Claim Interpretation

3. The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

4. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do

not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

5. This application includes one or more claim limitations that do not use the word "means," but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitation is: the "gait profiler" recited in **Claims 1-7 and 10-16**.

Because this claim limitation is being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it is being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

If applicant does not intend to have this limitation interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation recites sufficient structure to perform the claimed function so as to avoid it being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 101

6. 35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

7. Claims 1-7 and 10-16 are rejected under 35 U.S.C. 101 because the claimed invention is directed to a judicial exception (i.e., a law of nature, a natural phenomenon, or an abstract idea) without significantly more.

Step 1

Claims 1-7 and 10-16 are within the four statutory categories. Claims 1-7 are drawn to a method for generating a physiological assessment of a user, which is within the four statutory categories (i.e. process). **Claims 10-16** are drawn to a system for generating a physiological assessment of a user, which is within the four statutory categories (i.e. machine).

Prong 1 of Step 2A

Claim 1, which is representative of the inventive concept, recites: ***A method for generating a physiological assessment of a user from biomechanical data gathered from the user by a system having one or more processors, at least one orthotic device adapted to be worn by the user, at least one biomechanical sensor on each of the at least one orthotic device, and a gait profiler coupled to the one or more processors, the method comprising the steps of:***

- ***acquiring biomechanical data of the user to provide mechanical and biomechanical information associated to the user with the biomechanical sensors;***
- ***generating trajectory data of the user based on the acquired biomechanical data using the gait profiler and biomechanical sensors;***
- ***classifying the trajectory data into variables of interest data according to a set of key biomechanical features, wherein the variables of interest data are processed biomechanical data associated with specific times, postures, and/or activities, wherein the key***

biomechanical features are obtained by forming an indexed combination of curated data from a reference database and an outcome database;

- ***rejecting variables of interest data that do not fit a pre-determined acceptance criteria by validating a magnitude, shape, or trend over time of the trajectory data based on physiological determinants associated with a specific variable of interest;***
- ***extracting key features of the non-rejected variables of interest data;***
- ***calculating a current cumulative risk data based on the extracted key features, a user profile covariate data and the variables of interest data; and***
- ***generating the physiological assessment of the user using the current cumulative risk data.***

The underlined limitations as shown above, given the broadest reasonable interpretation, **cover the abstract idea of a mental process and/or a certain method of organizing human activity** because they recite a process that could be practically performed in the human mind (i.e. observations, evaluations, judgments, and/or opinions – in this case, the steps of acquiring biomechanical data, generating trajectory data, classifying the trajectory data, rejecting variables that do not fit a pre-determined acceptance criteria, extracting key features, calculating a current cumulative risk data, and generating the physiological assessment include at least evaluations and/or collecting information, analyzing it, and displaying certain results of the collection and analysis) or using a pen and paper, but for the recitation of generic computer components (i.e. the processors and gait profiler)/a computer environment (i.e. the orthotic device in communication with the processors and gait profiler)/the use of a computer (i.e. the processors and gait profiler) as a tool to perform the mental process, and/or managing personal behavior or relationships or interactions between people (i.e. social activities, teaching, and/or following rules or instructions

– in this case, the steps of acquiring biomechanical data, generating trajectory data, classifying the trajectory data, rejecting variables that do not fit a pre-determined acceptance criteria, extracting key features, calculating a current cumulative risk data, and generating the physiological assessment include following rules or instructions for assessing a patient's physiological state and notifying a user of the patient condition), e.g. see MPEP 2106.04(a)(2). Any limitations not identified above as part of the abstract idea are deemed "additional elements," and will be discussed in further detail below.

Furthermore, the abstract idea for **Claim 10** is identical as the abstract idea for **Claim 1**, because the only difference between **Claims 1 and 10** is that **Claim 1** recites a method, whereas **Claim 10** recites a system and biomechanical sensors utilized to obtain the biomechanical data.

Dependent **Claims 2-7 and 11-16** include other limitations, for example **Claims 2-3** recite criteria for rejecting variables of interest that do not fit pre-determined acceptance criteria, **Claims 4 and 13** recite adding additional biomechanical data to the assessment, **Claims 5 and 14** recite types of data utilized in obtaining key biomechanical features, **Claim 6** recites obtaining the biomechanical data from biomechanical sensors, **Claims 7 and 15-16** recite types of biomechanical sensors and the orientations of the sensors, **Claim 11** recites sorting and filtering the trajectory data, and **Claim 12** recites formatting the risk data for presentation to the user, but these only serve to further narrow the abstract idea, and a claim may not preempt abstract ideas, even if the judicial exception is narrow, e.g. see MPEP 2106.04, and/or do not further narrow the abstract idea and instead only recite additional elements, which will be further addressed below. Hence dependent **Claims 2-7 and 11-16** are nonetheless directed towards fundamentally the same abstract idea as independent **Claims 1 and 10**.

Prong 2 of Step 2A

Claims 1 and 10 are not integrated into a practical application because the additional elements (i.e. the non-underlined limitations above – in this case, the processors, the databases, the biomechanical sensors, the orthotic device, and the specific types of data processed by the aforementioned limitations) amount to no more than limitations which:

- amount to mere instructions to apply an exception – for example, the recitation of the processors, the databases, the gait profiler, the biomechanical sensors, and the orthotic device, which amounts to merely invoking a computer as a *tool* to perform the abstract idea, e.g. see [0036] and [0046]-[0047] of the present Specification, see MPEP 2106.05(f); and/or
- generally link the abstract idea to a particular technological environment or field of use – for example, the claim language of biomechanical data utilized to generate a physiological assessment for the user, which amounts to limiting the abstract idea to the field of healthcare/patient monitoring, see MPEP 2106.05(h);

Additionally, dependent **Claims 2-7 and 11-16** include other limitations, but these limitations also amount to no more than mere instructions to apply an exception (e.g. the types of biomechanical sensors recited in dependent **Claims 6-7 and 15-16**), generally linking the abstract idea to a particular technological environment or field of use (e.g. the types of data recited in dependent **Claims 3-5 and 13-14**), adding insignificant extra-solution activity to the abstract idea (e.g. the formatting and presenting of the data recited in dependent **Claim 12**), and/or do not include any additional elements beyond those already recited in independent **Claims 1 and 10**, and hence also do not integrate the aforementioned abstract idea into a practical application.

Hence **Claims 1-7 and 10-16** do not include additional elements that integrate the judicial exception into a practical application.

Step 2B

Claims 1 and 10 do not include additional elements that are sufficient to amount to "significantly more" than the judicial exception because the additional elements (i.e. the non-underlined limitations above – in this case, the processors, the databases, the biomechanical sensors, the orthotic device, and the specific types of data processed by the aforementioned limitations), as stated above, are directed towards no more than limitations that amount to mere instructions to apply the exception, and/or generally link the abstract idea to a particular technological environment or field of use, wherein the additional elements comprise limitations which:

- amount to elements that have been recognized as well-understood, routine, and conventional activity in particular fields, as demonstrated by:
    - The present Specification expressly disclosing that the structural additional elements are well-understood, routine, and conventional in nature:
        - [0036] and [0046]-[0047] of the Specification discloses that the additional elements (i.e. the processors, the databases, the biomechanical sensors, and the orthotic device) comprise a plurality of different types of generic computing systems;

- Relevant court decisions: The following are examples of court decisions demonstrating well-understood, routine and conventional activities, e.g. see MPEP 2106.05(d)(II):
  - Performing repetitive calculations, e.g. see *Parker v. Flook*, and/or *Bancorp Services v. Sun Life* – similarly, the current invention performs basic calculations (i.e. calculating a risk) and does not impose meaningful limits on the scope of the claims;
  - Electronic recordkeeping, e.g. see *Alice Corp v. CLS Bank* – similarly, the additional elements merely recite the creating and maintaining of curated data at the reference database and outcome database; and/or
  - Storing and retrieving information in memory, e.g. see *Versata Dev. Group, Inc. v. SAP Am., Inc.* – similarly, the current invention recites storing the curated data at the reference database and outcome database, and retrieving the curated data from storage in order to classify the trajectory data and ultimately generate the physiological assessment.

Dependent **Claims 2-7 and 11-16** include other limitations, but none of these limitations are deemed significantly more than the abstract idea because the additional elements recited in the aforementioned dependent claims similarly amount to mere instructions to apply the exception (e.g. the types of biomechanical sensors recited in dependent **Claims 6-7 and 15-16**), generally link the abstract idea to a particular technological environment or field of use (e.g. the types of data recited in dependent **Claims 4-5 and 13-14**), electronic recordkeeping (e.g. the

formatting and presenting of the assessment to the user recited in dependent **Claim 12**), and/or the limitations recited by the dependent claims do not recite any additional elements not already recited in independent **Claims 1 and 10**, and hence do not amount to "significantly more" than the abstract idea.

Hence, **Claims 1-7 and 10-16** do not include any additional elements that amount to "significantly more" than the judicial exception.

Thus, taken alone, the additional elements do not amount to significantly more than the abstract idea identified above. Furthermore, looking at the limitations as an ordered combination adds nothing that is not already present when looking at the elements taken individually, and there is no indication that the combination of elements improves the functioning of a computer or improves any other technology, and their collective functions merely provide conventional computer implementation.

Therefore, whether taken individually or as an ordered combination, **Claims 1-7 and 10-16** are nonetheless rejected under 35 U.S.C. 101 as being directed to non-statutory subject matter.

Claim Rejections - 35 USC § 103

8. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

9. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

10. Claims 1, 4-7, and 11-16 are rejected under 35 U.S.C. 103 as being unpatentable over Chang (US 2018/0177436) in view of Harper (US 2016/0333411).

11. Regarding **Claim 1**, Chang teaches the following: ***A method for generating a physiological assessment of a user from biomechanical data gathered from the user by a system having one or more processors*** (The system includes a processor, e.g. see Chang [0028].)***, at least one orthotic device adapted to be worn by the user*** (The system includes a wearable biomechanical sensing device, e.g. see Chang [0027].)***, at least one biomechanical sensor on each of the at least one orthotic device*** (The biomechanical sensing device includes various sensors to obtain various biomechanical metrics such as motion data, e.g. see Chang [0028] and [0030].)***, and a gait profiler coupled to the one or more processors*** (The system includes biomechanical processing modules to characterize gait dynamics, e.g. see Chang [0039].)***, the method comprising the steps of:***

- ***acquiring biomechanical data of the user to provide mechanical and biomechanical information associated to the user with the biomechanical sensors*** (The system includes a biomechanical sensing device (i.e. an orthotic device) including sensors that obtain data including gait metrics and activity data for a patient (i.e. mechanical and biomechanical information of a user), e.g. see Chang [0023]-[0025].)***;***

- ***generating trajectory data of the user based on the acquired biomechanical data using the gait profiler and biomechanical sensors*** (The system analyzes the patient gait metrics obtained from the biomechanical sensing device and generates various types of data, for example locomotion biomechanical measurements, step length asymmetries, shuffle detection, and tremors (i.e. any of which may be interpreted as trajectory data) for the user, e.g. see Chang [0095]-[0096] and [0102].)***;***

- ***classifying the trajectory data into variables of interest data according to a set of key biomechanical features, wherein the variables of interest data are processed biomechanical data associated with specific times, postures, and/or activities*** (The system determines (i.e. classifies) which data should be utilized in the generation of the locomotion biomechanical measurements (i.e. the data utilized is interpreted as "variables of interest"), step length asymmetries, and shuffle detection, and tremors (i.e. any of which may be interpreted as trajectory data), based on (i.e. according to) a set of thresholds (i.e. key biomechanical features), e.g. see Chang [0070], [0083], [0096], and [0112]. Furthermore, the data utilized may include data associated with limping, walking, and/or a risk of falling (i.e. data associated with posture and/or activities), e.g. see Chang [0083], [0096], and [0112], wherein the data associated with walking may include a step cadence based on a right and left ground contact time (i.e. specific times), e.g. see Chang [0071]-[0073].)***;***

- ***rejecting variables of interest data that do not fit a pre-determined acceptance criteria by validating a magnitude, shape, or trend over time of the trajectory data based on physiological determinants associated with a specific variable of interest*** (The system filters (i.e. rejects) data that does not fit criteria (i.e. not fitting a pre-determined acceptance criteria), for example data not satisfying a threshold may be classified as not a shuffle and/or not a tremor

(i.e. validating the magnitude or trend over time of the trajectory data), e.g. see Chang [0060], [0070], [0083], [0096], and [0112].)*;*

- ***extracting key features of the non-rejected variables of interest data*** (The system identifies indicators of risk (i.e. key features) from the non-filtered data, e.g. see Chang [0047], [0066], and [0106]-[0109].)*;*

- ***calculating a current cumulative risk data based on the extracted key features, a user profile covariate data and the variables of interest data*** (The system utilizes the indicators of risk (i.e. the key features), biometric data (i.e. user profile covariate data), and the non-filtered data (i.e. the variables of interest data) to determine a risk of a user action, for example a fall (i.e. cumulative risk data), e.g. see Chang [0066] and [0106]-[0109].)*; and*

- ***generating the physiological assessment of the user using the current cumulative risk data*** (The system performs a fall risk assessment that may trigger a response, for example providing the user with an alert, e.g. see Chang [0054], [0108], and [0132], Figs. 2 and 7.).

But Chang does not teach and Harper teaches the following:

- ***wherein the key biomechanical features are obtained by forming an indexed combination of curated data from a reference database and an outcome database*** (The system determines criteria for significant and scientific validity for data utilizing various sources, for example SNPedia, NCBI Variation Database (dsSNP), and peer reviewed literature, e.g. see Harper [0207]-[0219] – that is, the data is determined to be significant data (i.e. a key feature) based on analysis of the sources.)

Furthermore, before the effective filing date, it would have been obvious to one ordinarily skilled in the art of patient monitoring to modify Chang to incorporate utilizing the

results data from various sources including peer reviewed literature and databases to determine significant data as taught by Harper in order to support the determination made by the system with reliable data, e.g. see Harper [0219].

12. Regarding **Claim 4**, the combination of Chang and Harper teaches the limitations of **Claim 3**, and Chang further teaches the following:

- ***The method of claim 3, further comprising the step of adding biomechanically derived information to the assessment of the user*** (The data used in assessing the risk of a user action (i.e. the assessment of the user) includes data that is determined (i.e. derived) from raw sensor data, e.g. see Chang [0035], [0052], [0054], [0108], and [0132], Figs. 2 and 7.).

13. Regarding **Claim 5**, the combination of Chang and Harper teaches the limitations of **Claim 1**, and Harper further teaches the following: ***The method of claim 1,***

- ***wherein the reference database contains key biomechanical measurements and associated physiological determinant factors, and being constructed using peer-reviewed academic publications*** (The sources include peer reviewed literature including data regarding factors/words and phrases that reveals information that can affect or influence the ability to achieve an ideal weight, e.g. see Harper [0208]-[0209].)***; and***

- ***the outcome database contains linked key biomechanical measurements and associated statistically established outcomes, and is constructed based on user monitoring and experimentation*** (The sources include factors (i.e. key measurements) and results (i.e. outcomes) published in peer reviewed journals, e.g. see Harper [0208]-[0209].).

Furthermore, before the effective filing date, it would have been obvious to one ordinarily skilled in the art of patient monitoring to modify Chang to incorporate utilizing the results data from various sources including peer reviewed literature to determine significant data as taught by Harper in order to support the determination made by the system with reliable data, e.g. see Harper [0219].

14. Regarding **Claim 6**, the combination of Chang and Harper teaches the limitations of **Claim 1**, and Chang further teaches the following:

- ***The method of claim 1, wherein the biomechanical data of the user is acquired from biomechanical sensors positioned on the user*** (The data is captured from biomechanical sensing devices positioned on the body of a user, e.g. see Chang [0027]-[0030].).

15. Regarding **Claim 7**, the combination of Chang and Harper teaches the limitations of **Claim 6**, and Chang further teaches the following:

- ***The method of claim 6, wherein the biomechanical sensors include inertial and angular sensors positioned on a lower-body orthotic device worn by the user*** (The biomechanical sensing devices may be positioned on a user's lower leg, and may include an inertial measurement unit and functionality that measures angular orientation, e.g. see Chang [0027]-[0031].).

16. Regarding **Claim 10**, the limitations of **Claim 10** are substantially similar to those claimed in **Claim 1**, with the sole difference being that **Claim 1** recites a method whereas **Claim 10** recites a system including biomechanical sensors and a processor and memory in

communication with the biomechanical sensors. Specifically pertaining to **Claim 10**, Examiner notes that Chang teaches a plurality of biomechanical sensing devices in communication with a processor and a memory, e.g. see Chang [0028], and hence the grounds of rejection provided above for **Claim 1** are similarly applied to **Claim 10**.

17. Regarding **Claim 11**, the combination of Chang and Harper teaches the limitations of **Claim 10**, and Chang further teaches the following: ***The system of claim 10, wherein the memory comprises further instructions stored therein that when executed on the processor further perform the steps of:***

- ***sorting and labeling the trajectory data into discrete segments*** (The system may sort the data into step-wise window segments, e.g. see Chang [0070].)***; and***
- ***filtering the discrete segments of trajectory data so as to reduce a number of individual frames and remove noise*** (The segmented data may be analyzed to determine which portions satisfy various conditions/thresholds, for example a minimum amplitude, wherein the data that does not satisfy the conditions is filtered, e.g. see Chang [0060], [0070], [0083], [0096], and [0112].)***;***
- ***wherein the step of classifying the trajectory data into variables of interest data according to a set of key biomechanical features is performed on the filtered discrete segments of trajectory data*** (The data that is not filtered is considered to be usable for determining a user condition, for example a state of mobility including a limp, e.g. see Chang [0060], [0070], [0083], [0096], and [0112].)**.**

18. Regarding **Claim 12**, the combination of Chang and Harper teaches the limitations of **Claim 10**, and Chang further teaches the following:

- ***The system of claims claim 10, wherein the memory comprises further instructions stored therein that when executed on the processor further perform the steps of formatting and simplifying the risk data of the assessment of the user for presentation to the user*** (The assessed risk for a user action (i.e. the assessment of the user based on the risk data) may be utilized to construct an alert that is displayed to the user, e.g. see Chang [0054], [0108], and [0132], Figs. 2 and 7.).

19. Regarding **Claim 14**, the combination of Chang and Harper teaches the limitations of **Claim 10**, and Harper further teaches the following: ***The system of claim 10, wherein:***

- ***the reference database contains key biomechanical measurements and associated physiological determinant factors, from peer-reviewed academic publications*** (The sources include peer reviewed literature including data regarding factors/words and phrases that reveals information that can affect or influence the ability to achieve an ideal weight, e.g. see Harper [0208]-[0209].)***; and***

- ***the outcome database contains linked key biomechanical measurements and associated statistically established outcomes, from user monitoring and experimentation*** (The sources include factors (i.e. key measurements) and results (i.e. outcomes) published in peer reviewed journals, e.g. see Harper [0208]-[0209].).

Furthermore, before the effective filing date, it would have been obvious to one ordinarily skilled in the art of patient monitoring to modify Chang to incorporate utilizing the results data from various sources including peer reviewed literature to determine significant data

as taught by Harper in order to support the determination made by the system with reliable data, e.g. see Harper [0219].

20. Regarding **Claims 13 and 15-16**, the limitations of **Claims 13 and 15-16** are substantially similar to those claimed in **Claims 4 and 6-7**, with the sole difference being that **Claims 4 and 6-7** recite a method whereas **Claims 13 and 15-16** recite a system including biomechanical sensors and a processor and memory in communication with the biomechanical sensors. Specifically pertaining to **Claims 13 and 15-16**, Examiner notes that Chang teaches a plurality of biomechanical sensing devices in communication with a processor and a memory, e.g. see Chang [0028], and hence the grounds of rejection provided above for **Claims 4 and 6-7** are similarly applied to **Claims 13 and 15-16**.

Subject Matter Free From Prior Art

21. **Claims 2-3** are not presently rejected under 35 U.S.C. 102 or 103, and hence would be in condition for allowance if amended to overcome the rejections presented under 35 U.S.C. 101. The following represents Examiner's characterization of the most relevant prior art references and the differences between the present claim language and the prior art references in view of 35 U.S.C. 102 and/or 103:

With regards to 35 U.S.C. 102 and/or 103, the following represents the closest prior art to the claimed invention, as well as the differences between the prior art and the limitations of the presently claimed invention.

As shown above, Chang (US 2018/0177436) teaches a system for monitoring patient gait and movement that identifies high risk patients based on comparing the monitored parameters to

various criteria and thresholds, for example data defining movement as a shuffle and/or a tremor, and outputting a report indicating the patient risk.

Harper (US 2016/0333411) teaches a system for evaluating patient parameters in order to generate a personalized health profile utilizing various criteria including peer-reviewed journals and human studies.

Giuffrida (US 2013/0123666) teaches a system for monitoring patient gait and movement and analyzing the monitored data using pre-determined thresholds that are stored as a training set and/or in a database.

However, Chang, Harper, and Giuffrida do not teach that the step of rejecting variables of interest data comprises flagging an increase in double support time in response to a statistically significant increase in double-support phase of walking for consecutive months, as is claimed by dependent **Claim 2**. Furthermore, Chang, Harper, and Giuffrida do not teach that the step of rejecting variables of interest data comprises flagging an increase in double support time in response to a mean increase in double support time of more than 1% of gait cycle, as is claimed by dependent **Claim 3**.

The aforementioned references are understood to be the closest prior art. Various aspects of the claimed invention are known individually, but for the reasons disclosed above, the particular manner in which the elements of the present invention are claimed, when considered as an ordered combination, distinguishes from the aforementioned references and hence the invention recited in **Claims 2-3** is not considered to be disclosed by and/or obvious in view of the inventions of the closest prior art references.

Response to Arguments

22. Applicant's arguments, see Remarks, filed June 4, 2026, with respect to rejections of **Claims 1-7 and 10-16** under 35 U.S.C. 112(a) have been fully considered and, in combination with the amendments to the Claims, are persuasive. The rejections of **Claims 1-7 and 10-16** under 35 U.S.C. 112(a) have been withdrawn.

23. Applicant's arguments, see Remarks, filed June 4, 2026, with respect to the rejections of **Claims 1-7 and 10-16** under 35 U.S.C. 101 have been fully considered but are not persuasive.

Applicant alleges that the claimed invention is patent eligible because it is not directed towards an abstract idea, specifically because it is directed towards a technical improvement in orthotic monitoring, e.g. see pgs. 11-12 of Remarks – Examiner disagrees.

Applicant alleges that the claimed limitations recite a "physiological gatekeeper" that is distinguished from a "generic outlier filter," but the claim language recites "rejecting variables that do not fit a pre-determined acceptance criteria by validating a magnitude, shape, or trend over time of the trajectory data based on of physiological determinants associated with a specific variable of interest." That is, given the broadest reasonable interpretation, the claim language recites filtering physiological data based on any one of magnitude, shape, or a trend over time. As shown above, this feature is part of both a mental process and/or a certain method of managing personal behavior because this feature corresponds to at least the analysis portion of collecting data, analyzing it, and displaying certain results of the analysis, and/or comprises following rules or instructions to ultimately generate a patient physiological assessment.

Applicant further alleges that the claimed invention is patent eligible because the rejection logic is not conventional signal processing, e.g. see pgs. 12-13 of Remarks – Examiner disagrees.

Initially, Examiner notes that the Claims being narrowly claimed is not dispositive in determining the eligibility of the Claims. The Court has held that a claim may not preempt abstract ideas, laws of nature, or natural phenomena, even if the judicial exception is narrow, e.g. see MPEP 2106.04. That is, a claim reciting a narrow abstract idea nonetheless recites an abstract idea, and must be evaluated under the remainder of the requirements under 35 U.S.C. 101.

Additionally, "the novelty of any element or steps in a process, or even of the process itself, is of ***no relevance*** in determining whether the subject matter of a claim falls within the 101 categories of possibly patentable subject matter," and specifically, a finding of a lack of novelty under 35 U.S.C. 102 or obviousness under 35 U.S.C. 103 of a claimed invention does not necessarily indicate that claimed invention is therefore patent eligible. Because they are separate and distinct requirements from eligibility, patentability of the claimed invention under 35 U.S.C. 102 and 103 with respect to the prior art is neither required for, nor a guarantee of, patent eligibility under 35 U.S.C. 101, e.g. see MPEP 2106.05(I). That is, even assuming, *arguendo*, that the claim limitations recite a specific type of data that is not anticipated by and/or rendered obvious in view of prior art under 35 U.S.C. 102 and/or 103, this consideration is irrelevant to the determination of subject matter eligibility. Furthermore, Examiner further notes, as will be explained in further detail below with regards to 35 U.S.C. 103, that the filtering of Chang does indeed teach the step of rejecting variables as is presently claimed in **Claims 1 and 10**.

Additionally, [0037] of the as-filed Specification discloses that "existing technology…allow for discrete observations over time, however **they do not have sufficient sensors to track multiple body segments**" (emphasis added). That is, [0037] of the as-filed Specification discloses that the problem in existing technology addressed by the claimed invention is insufficient sensors. However, the claim language does not claim any particular number and/or type of specialized sensor to obtain the data, any type of unique methodology for gathering the data, and/or any particular configuration of the sensors, and hence there is an insufficient nexus between the claimed limitations and the alleged technical improvement.

For the aforementioned reasons, **Claims 1-7 and 10-16** are rejected under 35 U.S.C. 101.

24. Applicant's arguments, see Remarks, filed June 4, 2026, with respect to the rejections of **Claims 1-7 and 10-16** under 35 U.S.C. 103 have been fully considered but are not persuasive.

Applicants first allege that Chang and Harper do not teach rejecting data based on physiological authenticity, specifically the magnitude, shape, or trend over time of trajectory data, e.g. see pg. 14 of Remarks – Examiner disagrees.

Initially, Examiner notes that "trajectory data" is defined by **Claims 1 and 10** as data that is generated based on acquired biomechanical data using the gait profiler and the biomechanical sensors. Furthermore, [0054] of the as-filed Specification discloses that the trajectory data may include "3D acceleration data of body segments and/or angular data from joints." That is, given the broadest reasonable interpretation in view of the Claims and the Specification, "trajectory data" may be interpreted as any type of data derived from the biomechanical data using the gait profiler and the biomechanical sensors, with examples of "trajectory data" being 3D acceleration data of body segments and/or angular data of joints.

[0057] of Chang teaches that sensor data includes kinematic data relating to motion and/or orientation of some portion of a user's body, and [0060] of Chang teaches that kinematic data may be pre-processed, for example by filtering the data. Furthermore, the system generates locomotion biomechanical measurements by segmenting the kinematic data, wherein the segmenting can be performed by "counting extrema exceeding a minimum amplitude requirement in the filtered, three-dimensional acceleration magnitude as measured by the sensor," e.g. see Chang [0070]. That is, Chang teaches measuring a magnitude of acceleration and counting a number of extrema exceeding a minimum amplitude to generate locomotion biomechanical measurements. Hence locomotion biomechanical measurements are properly interpreted as "trajectory data" in view of the Claim language of **Claims 1 and 10** and the disclosures of the as-filed Specification. Therefore, Chang teaches using at least a magnitude of trajectory data as a basis for the filtering (i.e. rejection) of data and is not deficient to teach this feature.

Applicant further alleges that Harper is not properly combined with Chang because it is non-analogous art and because there is no motivation to combine the references, e.g. see pgs. 14-15 of Remarks – Examiner disagrees.

In response to applicant's argument that Harper is nonanalogous art, it has been held that a prior art reference must either be in the field of the inventor's endeavor or, if not, then be reasonably pertinent to the particular problem with which the inventor was concerned, in order to be relied upon as a basis for rejection of the claimed invention. See *In re Oetiker*, 977 F.2d 1443, 24 USPQ2d 1443 (Fed. Cir. 1992). In this case, the claimed invention concerns the field of biomechanical observations (i.e. patient monitoring), e.g. see [0037] of the as-filed Specification, the invention of Chang concerns biomechanical monitoring, e.g. see Chang

[0005], and the invention of Harper concerns monitoring a subject while underdoing treatment to evaluate the efficacy of the treatment, e.g. see Harper [0094], and hence Chang and Harper are in the same field of present application. Additionally, [0037] of the as-filed Specification discloses that the claimed invention addresses the problem of measuring patient parameters to "reveal key details of general health status and disease progression." Similarly, the invention of Chang addresses the problem of tracking mobility progress and assisting or directing a patient for mobility improvement, e.g. see Chang [0016], and Harper addresses the need for "methods to determine a subject's overall weight profile and propensity toward deleterious weight changes…so that steps can be taken to prevent, reduce or inhibit weight conditions, disorders, or diseases." Hence, Chang and Harper are also reasonably pertinent to the particular problem with which the inventor was concerned, and are both properly considered analogous art.

In response to applicant's argument that there is no teaching, suggestion, or motivation to combine the references, the examiner recognizes that obviousness may be established by combining or modifying the teachings of the prior art to produce the claimed invention where there is some teaching, suggestion, or motivation to do so found either in the references themselves or in the knowledge generally available to one of ordinary skill in the art. See *In re Fine*, 837 F.2d 1071, 5 USPQ2d 1596 (Fed. Cir. 1988), *In re Jones*, 958 F.2d 347, 21 USPQ2d 1941 (Fed. Cir. 1992), and *KSR International Co. v. Teleflex, Inc.*, 550 U.S. 398, 82 USPQ2d 1385 (2007). In this case, Chang does not specifically teach utilizing an indexed combination of curated data from a reference databased and an outcome database, but Harper teaches that utilizing data from multiple databases and/or sources enables the processing operations to provide reliable, relevant, valid, and/or significant data, e.g. see Harper [0210] and [0219]. Hence, it would have been obvious to modify Chang to incorporate the multiple databases and/or

sources including peer reviewed academic publications in order to enable the processing operations to provide reliable, relevant, valid, and/or significant data, e.g. see Harper [0210] and [0219].

For the aforementioned reasons, **Claims 1-7 and 10-16** are rejected under 35 U.S.C. 103.

Conclusion

25. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure is as follows:

- Jain (US 2014/0371886) – teaches a system for monitoring user athletic performance including a database that stores archived user data and threshold values for evaluating the user data.

26. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

27. Any inquiry concerning this communication or earlier communications from the examiner should be directed to JOHN P GO whose telephone number is (703)756-1965. The examiner can normally be reached Monday-Friday 9am-6pm Pacific.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, PETER H CHOI can be reached at (469)295-9171. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/JOHN P GO/
Primary Examiner, Art Unit 3681